2ND EXPERIMENT

```
mysql> CREATE DATABASE COLLEGE2;
Query OK, 1 row affected (3.53 sec)

mysql>
mysql> USE COLLEGE2;
Database changed
mysql> CREATE TABLE STUDENT2
    -> (
    ->     RegNo INT,
    ->     S1Name VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> DESC STUDENT2;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| RegNo  | int         | YES  |     | NULL    |       |
| S1Name | varchar(15) | YES  |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
2 rows in set (0.01 sec)

mysql> CREATE TABLE STUDENT3
    -> (
    ->     RegNo INT
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DESC STUDENT3;
+-------+------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+-------+------+------+-----+---------+-------+
| RegNo | int  | YES  |     | NULL    |       |
+-------+------+------+-----+---------+-------+
1 row in set (0.00 sec)
```

```
mysql> DESC FACULTY;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| FacNo    | varchar(4)  | YES  |     | NULL    |       |
| FacName  | varchar(15) | YES  |     | NULL    |       |
| Gender   | char(1)     | YES  |     | NULL    |       |
| DOB      | date        | YES  |     | NULL    |       |
| DOJ      | date        | YES  |     | NULL    |       |
| MobileNo | varchar(10) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)

mysql> CREATE TABLE DEPARTMENT
    -> (
    ->     DeptNo VARCHAR(4),
    ->     DeptName VARCHAR(15),
    ->     DeptHead VARCHAR(4)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DESC DEPARTMENT;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| DeptNo   | varchar(4)  | YES  |     | NULL    |       |
| DeptName | varchar(15) | YES  |     | NULL    |       |
| DeptHead | varchar(4)  | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE COURSE
    -> (
    ->     CourseNo VARCHAR(3),
    ->     CourseDesc VARCHAR(14),
    ->     CourseType CHAR(1),
    ->     SemNo CHAR(1),
    ->     HallNo VARCHAR(4),
    ->     FacNo VARCHAR(4)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DESC COURSE;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| CourseNo   | varchar(3)  | YES  |     | NULL    |       |
| CourseDesc | varchar(14) | YES  |     | NULL    |       |
| CourseType | char(1)     | YES  |     | NULL    |       |
| SemNo      | char(1)     | YES  |     | NULL    |       |
| HallNo     | varchar(4)  | YES  |     | NULL    |       |
| FacNo      | varchar(4)  | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
6 rows in set (0.01 sec)

mysql> ALTER TABLE STUDENT2
    -> ADD PRIMARY KEY (RegNo);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

QUESTIONS AND ITS ANSWERS

1.

```
mysql> ALTER TABLE STUDENT2
    -> ADD PRIMARY KEY (RegNo);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> DESC STUDENT2;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| RegNo  | int         | NO   | PRI | NULL    |       |
| S1Name | varchar(15) | YES  |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

2.

```
mysql> ALTER TABLE FACULTY
    -> ADD PRIMARY KEY (FacNo);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE FACULTY
    -> ADD CONSTRAINT chk_faculty_gender
    -> CHECK (Gender IN ('M','F'));
Query OK, 0 rows affected (0.13 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC FACULTY;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| FacNo    | varchar(4)  | NO   | PRI | NULL    |       |
| FacName  | varchar(15) | YES  |     | NULL    |       |
| Gender   | char(1)     | YES  |     | NULL    |       |
| DOB      | date        | YES  |     | NULL    |       |
| DOJ      | date        | YES  |     | NULL    |       |
| MobileNo | varchar(10) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

3.

```
mysql> ALTER TABLE DEPARTMENT
    -> ADD PRIMARY KEY (DeptNo);
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC DEPARTMENT;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| DeptNo   | varchar(4)  | NO   | PRI | NULL    |       |
| DeptName | varchar(15) | YES  |     | NULL    |       |
| DeptHead | varchar(4)  | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

4.

```
mysql> ALTER TABLE COURSE
    -> ADD PRIMARY KEY (CourseNo);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE COURSE
    -> ADD CONSTRAINT chk_semno
    -> CHECK (SemNo BETWEEN '1' AND '6');
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC COURSE;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| CourseNo   | varchar(3)  | NO   | PRI | NULL    |       |
| CourseDesc | varchar(14) | YES  |     | NULL    |       |
| CourseType | char(1)     | YES  |     | NULL    |       |
| SemNo      | char(1)     | YES  |     | NULL    |       |
| HallNo     | varchar(4)  | YES  |     | NULL    |       |
| FacNo      | varchar(4)  | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

5.

```
mysql> ALTER TABLE STUDENT2
    -> ADD CONSTRAINT uq_s1name
    -> UNIQUE (S1Name);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC STUDENT2;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| RegNo  | int         | NO   | PRI | NULL    |       |
| S1Name | varchar(15) | YES  | UNI | NULL    |       |
+--------+-------------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

6.

```
mysql> ALTER TABLE STUDENT3
    -> ADD CONSTRAINT fk_student_regno
    -> FOREIGN KEY (RegNo)
    -> REFERENCES STUDENT2(RegNo);
Query OK, 0 rows affected (0.13 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> DESC STUDENT3;
+-------+------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+-------+------+------+-----+---------+-------+
| RegNo | int  | YES  | MUL | NULL    |       |
+-------+------+------+-----+---------+-------+
1 row in set (0.00 sec)
```